

Change of basis

$T: V \longrightarrow V$ linear transformation

$B = (v_1, v_2, \dots, v_n)$: basis for V

$\downarrow$

$$U: F^n \xrightarrow{\cong} V$$

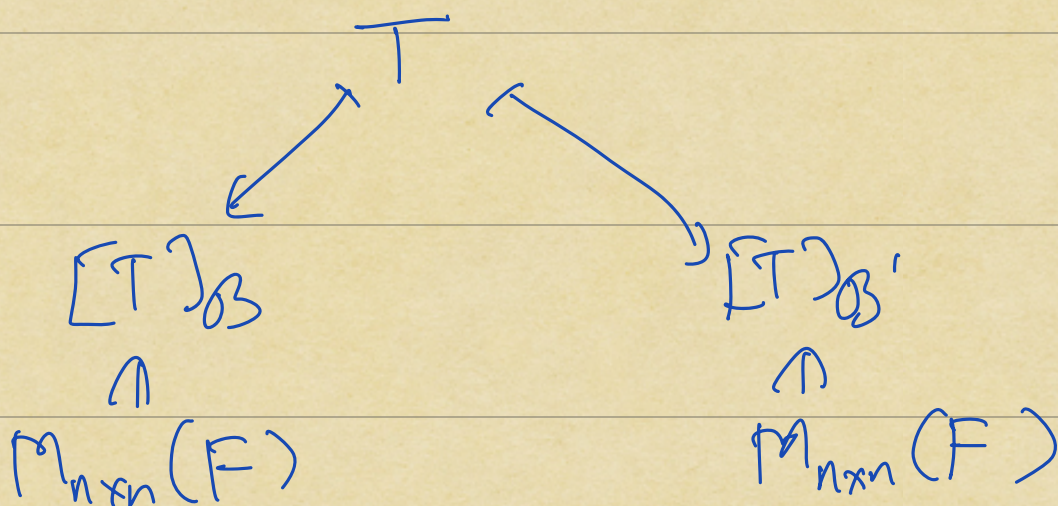
$e_i \longmapsto v_i$

$$\begin{array}{ccc} F^n & \xrightarrow[\cong]{U} & V \\ \downarrow U^{-1} \circ T \circ U & & \downarrow T \\ F^n & \xleftarrow[U^{-1}]{\cong} & V \end{array}$$

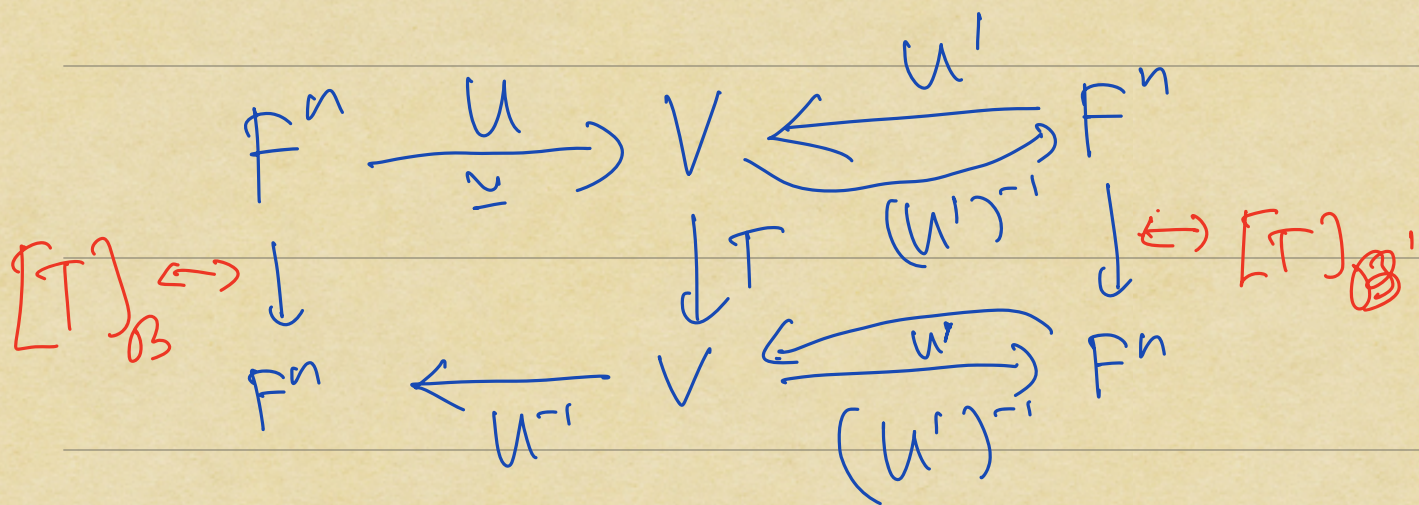
$U^{-1} \circ T \circ U$ is represented by
an $n \times n$ -matrix denoted by $[T]_B$

$$[T]_{\mathcal{B}, \mathcal{B}}$$

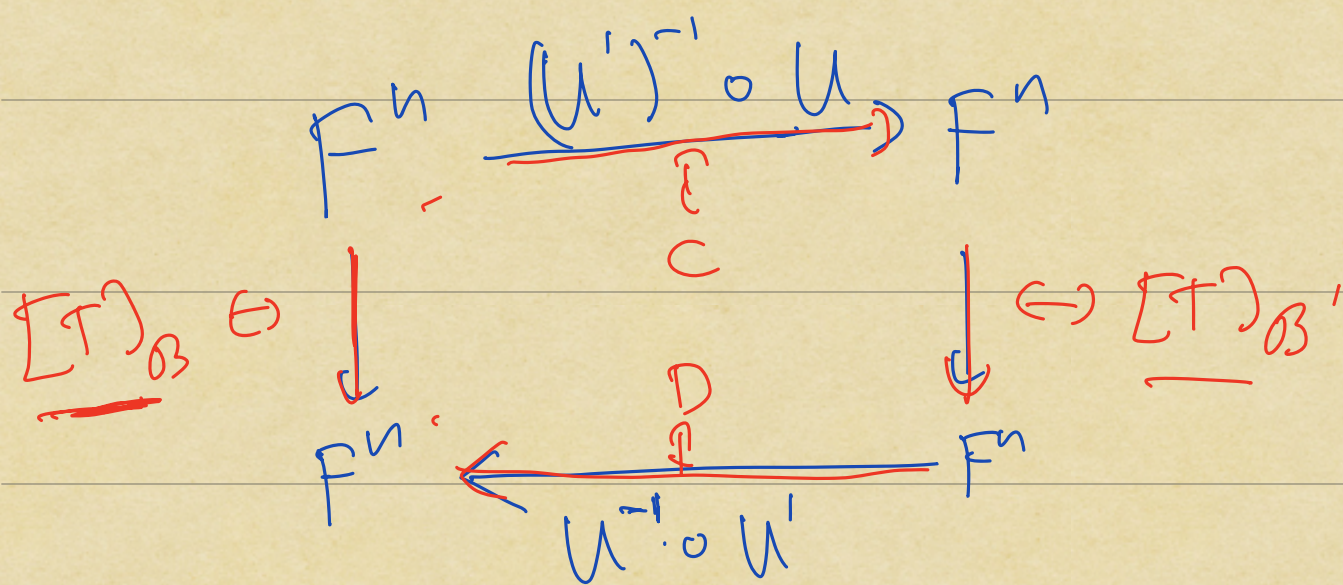
Suppose that $\mathcal{B}' = (v'_1, v'_2, \dots, v'_n)$ is a different choice of basis for V corresponding to an isomorphism $U: F^n \xrightarrow{\sim} V$

$$e_i \mapsto v'_i$$


Q: What is the relationship between $[T]_{\mathcal{B}}$ & $[T]_{\mathcal{B}'}$?



Extract the outer edges



Observation ①: $(U')^{-1} \circ U \Leftrightarrow C \in M_{n \times n}(F)$
 $U^{-1} \circ U' \Leftrightarrow D \in M_{n \times n}(F)$

$$[T]_B = D [T]_{B'} C$$

Defⁿ (a) I_n : identity matrix

$\uparrow$
 $M_{n \times n}(F)$ is given by

$$I_n = \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix} = [e_1, e_2, \dots, e_n]$$

e.g. $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

I_n is the matrix for the identity linear transformation

$$\text{id}_{F^n} : F^n \longrightarrow F^n.$$

(b) $C \in M_{n \times n}(F)$ is invertible

if the corresponding linear transformation

$$T : F^n \longrightarrow F^n \text{ is invertible}$$

In this case, the inverse matrix C^{-1} is the $n \times n$ -matrix for $T^{-1} : F^n \longrightarrow F^n$.

Observation (2)

$T \leftrightarrow C \in M_{n \times n}(F)$ with inverse C^{-1}

$$C^{-1}C = CC^{-1} = I_n$$

$$\begin{array}{ccc} \uparrow & & \uparrow \\ T^{-1} \circ T & = & T \circ T^{-1} = \text{id}_F^n \end{array}$$

Observation (3) (socks-then-shoes principle)

$$\bullet \quad V \xrightarrow[\cong]{T} W \xrightarrow[\cong]{S} U \quad \text{isomorphisms}$$

$S \circ T$ is also an isomorphism
with inverse $(S \circ T)^{-1} = T^{-1} \circ S^{-1}$

$$\begin{array}{ccc} V & \xrightarrow{S \circ T} & U \\ & \xleftarrow{T^{-1} \circ S^{-1}} & \end{array}$$

$\bullet \quad C_1, C_2 \in M_{n \times n}(F)$ invertible

Then $C_1 C_2$ is also invertible
and $(C_1 C_2)^{-1} = C_2^{-1} C_1^{-1}$

Observation (4)

$$\cdot (T^{-1})^{-1} = T$$

$$\cdot (C^{-1})^{-1} = C$$

Observation (1): $(U'^{-1} \circ U) \Leftrightarrow C \in M_{n \times n}(F)$
 $U'^{-1} \circ U' \Leftrightarrow D \in M_{n \times n}(F)$

$$[T]_B = D [T]_{B'} C$$

But actually $D = C^{-1}$

So we can rewrite the equality

as

$$\cdot [T]_B = C^{-1} [T]_{B'} C$$

$\hookrightarrow C(\cdot)C^{-1}$

$$\cdot C [T]_B C^{-1} = C (C^{-1} [T]_{B'} C) C^{-1}$$

Associativity $\rightarrow$!!

$$(C^{-1})[T]_{B'}(C^{-1})$$

$$\stackrel{\text{Identity}}{=} I_n [T]_{B'} I_n \\ = [T]_{B'}$$

$$[T]_{B'} = C [T]_B C^{-1} ; [T]_B = C^{-1} [T]_{B'} C$$

Remains to understand C .

$$C \leftrightarrow (U')^{-1} \circ U : F^n \longrightarrow F^n$$

$$U : e_i \mapsto v_i$$

$$U' : e_i \mapsto v_i' \leftrightarrow (U')^{-1} : \underline{v_i'} \mapsto e_i$$

$$\begin{aligned} ((U')^{-1} \circ U)(\underline{e_j}) &= (U')^{-1}(U(e_j)) \\ &= (U')^{-1}(\underline{v_j}) \end{aligned}$$

$$\begin{aligned} &= (U')^{-1}(\sum a_{i,j} v_i') \\ &= \sum a_{i,j} e_i \end{aligned}$$

$$C = \begin{bmatrix} a_{1,1} & a_{1,2} & \dots & a_{1,n} \\ a_{2,1} & \vdots & & \vdots \\ a_{3,1} & & & \vdots \\ \vdots & & & \vdots \\ a_{n,1} & \dots & \dots & a_{n,n} \end{bmatrix}$$

is the change of basis matrix
from B' to B

Properties of matrix multiplication

(1) (Identity)

$$T \Leftrightarrow C \in M_{n \times n}(F)$$

$$C \cdot I_n = C = I_n \cdot C$$

$$\begin{array}{c} \downarrow \quad \quad \downarrow \quad \quad \downarrow \\ T \circ \text{id}_F^n = T = \text{id}_F^n \circ T \end{array}$$

(2) (Associativity)

$$T_1, T_2, T_3 \Leftrightarrow C_1, C_2, C_3 \in M_{n \times n}(F)$$

$$C_1 (C_2 C_3) = (C_1 C_2) C_3$$

$$\begin{array}{c} \downarrow \quad \quad \downarrow \\ T_1 \circ (T_2 \circ T_3) = (T_1 \circ T_2) \circ T_3 \end{array}$$